ITM SKILLS UNIVERSITY

Department of Computer Science & Engineering

SOFTWARE ENGINEERING & PROJECT MANAGEMENT

PETPAL

Case Study No. 64

TEST PLAN & EVIDENCE

IEEE 829 Test Strategy, Planned Functional Test Cases, Boundary Value Analysis, Decision Tables & SQA Metrics

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SUBMITTED BY

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Project: PetPal — Appointment Booking and Records App for a Veterinary Clinic Chain
Case Study: No. 64
Subject: Software Engineering & Project Management

SECTION 1 & 2

1. Master Test Plan & 2. Planned Functional Test Cases Suite

TABLE 2.1: PLANNED FUNCTIONAL TEST CASES SUITE (TC-01 TO TC-16 SAMPLE)

ID	REQ ID	TEST SCENARIO	PRECONDITION	EXPECTED RESULT	STATUS
TC-01	FR-01	Pet Owner Registration	Email not in DB	Account created; password hashed with bcrypt	PLANNED
TC-03	FR-02	Create Pet Profile (Bruno)	User logged in	Profile saved and displayed in dashboard	PLANNED
TC-07	FR-05	Slot Availability Matrix	Vet selected	Returns 30-min available slots; disables booked	PLANNED
TC-08	FR-06	10-Min Temporary Slot Hold	Slot selected	Slot locked for 10 min during checkout	PLANNED
TC-11	FR-08	Payment Gateway Webhook	Hold active	Appt Confirmed; payment record stored; receipt issued	PLANNED
TC-13	FR-09	Booster Reminder Batch	Due date T+14d	Queues SMS/Email: 'Bruno due for Rabies'	PLANNED
TC-16	FR-12	Legacy Sync Boundary	Appt Confirmed	Payload contains ONLY schedule metadata; NO notes	PLANNED

SECTION 3 & 4

3. Boundary Value Analysis (BVA) & 4. Equivalence Partitioning

TABLE 3.1: 7-POINT BVA PET AGE BOUNDS (0.1 TO 30.0 YRS)

POINT	VALUE	CLASS	EXPECTED RESULT
Min - 1	0.0 yrs	Invalid	Reject (Age > 0)
Min	0.1 yrs	Valid	Accept (Puppy/Kitten)
Nominal	5.0 yrs	Valid	Accept (Standard)
Max	30.0 yrs	Valid	Accept (Geriatric)
Max + 1	30.1 yrs	Invalid	Reject (> 30 threshold)

TABLE 4.1: VACCINATION REMINDER DECISION TABLE

RULE	DUE?	COMPLETED?	CHANNEL?	SYSTEM ACTION
R1	YES	NO	YES	SEND REMINDER
R3	YES	YES	YES	SUPPRESS (DONE)
R5	YES	NO	NO	LOG MISSING CHANNEL
R9	OVERDUE	NO	YES	URGENT ALERT + MARK OVERDUE